

R2 and T1 limit the charging current if something fails or battery terminals get short-circuited accidentally.

To set a charging current, while a multimeter is connected in series with the battery and source supply is present, adjust potentiometer VR1 slowly until the charging current reaches its required value.

Overcharge and deep-discharge protection have been shown in dotted areas of the circuit diagram. All components in these areas are subjected to a maximum of the battery voltage and not the DC source voltage. This makes the circuit work under a wide range of source voltages and without any influence from the charging current value. Set overcharge and deep-discharge voltage of the battery using potentiometers VR1 and VR2 before charging the battery.

In overcharge protection, zener

diode ZD1 starts conducting after its breakdown voltage is reached, i.e., it conducts when the battery voltage goes beyond a prefixed high level. Adjust VR2 when the battery is fully charged (say, 13.5V in case of a 12V battery) so that V_{GS} of T5 is set to zero and hence charging current stops flowing to the battery. LED1 glows to indicate that the battery is fully charged. When LED1 glows, the internal LED of the optocoupler also glows and the internal transistor conducts. As a result, gate-source voltage (V_{GS}) of MOSFET T5 becomes zero and charging stops.

Normally, zener diode ZD2 conducts to drive transistor T3 into conduction and thus make transistor T4 cut-off. If the battery terminal voltage drops to, say, 11V in case of a 12V battery, adjust potentiometer VR3 such that transistor T3 is cut-off and T4 conducts.

LED2 will glow to indicate that the battery voltage is low.

Values of zener diodes ZD1 and ZD2 will be the same for 6V, 9V and 12V batteries. For other voltages, you need to suitably change the values of ZD1 and ZD2. Charging current provided by this circuit is 1 mA to 1 A, and no heat-sink is required for T5. If the maximum charging current required is 5A, put another LM236-5 in series with diode D2, change the value of R1 to 1 kilo-ohm, replace D1 with two SB560 devices in parallel and provide a good heat-sink for MOSFET T1. TO-220 package of IRF540 can handle up to 50W.

Assemble the circuit on a general-purpose PCB and enclose in a box after setting the charging current, overcharge voltage and deep-discharge voltage. Mount potentiometers VR1, VR2 and VR3 on the front panel of the box. ●

FREEZER MONITOR ALARM

■ T.K. HAREENDRAN

All items stored in a deep freezer will thaw out if, for some reason, the temperature inside the freezer rises to the thaw point. However, a freezer monitor alarm can warn you of the rising temperature before the thaw point is reached.

This simple circuit is powered by

regulated 9V and built around a few discrete components that are readily available in the market. Current consumption of the circuit is only a few milliamperes in idle state. So you can also use a 9V 6F22/PP3 type compact battery for powering the circuit.

Germanium transistor AC125 (T1) acts as the temperature sensor. For sensing, the monitor circuit uses the leakage current of the transistor, which rises with temperature, and is adjustable from -18°C to room temperature.

Working of the

circuit is simple. At normal temperature of the freezer, the resistance between the emitter and collector of transistor AC125 is very high and therefore the voltage at the gate terminal of SCR BT169 is too low to fire it. As a result, the rest of the circuit is disabled. Relay RL1 does not energise and hence the buzzer remains silent.

When the temperature of the freezer rises beyond the level fixed by preset VR1, the gate voltage of SCR BT169 (T2) goes high and the SCR is fired. Consequently, the gate terminal of MOSFET BS170 (T3) receives positive voltage through resistor R2 and relay RL1 energises to latch. The buzzer sounds to indicate higher temperature of the freezer, unless you switch off the circuit through switch S1. The relay contacts can also be used to control

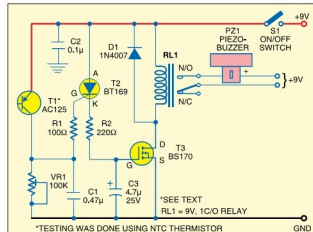


Fig. 1: Freezer alarm circuit

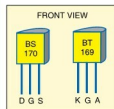
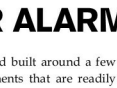


Fig. 2: Pin Configurations of BT169 and BS170